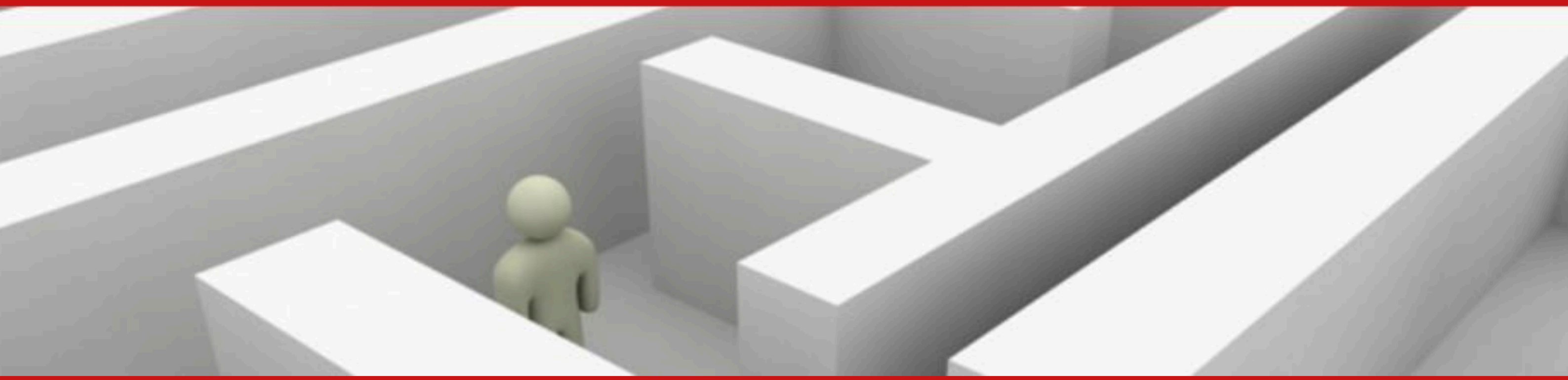
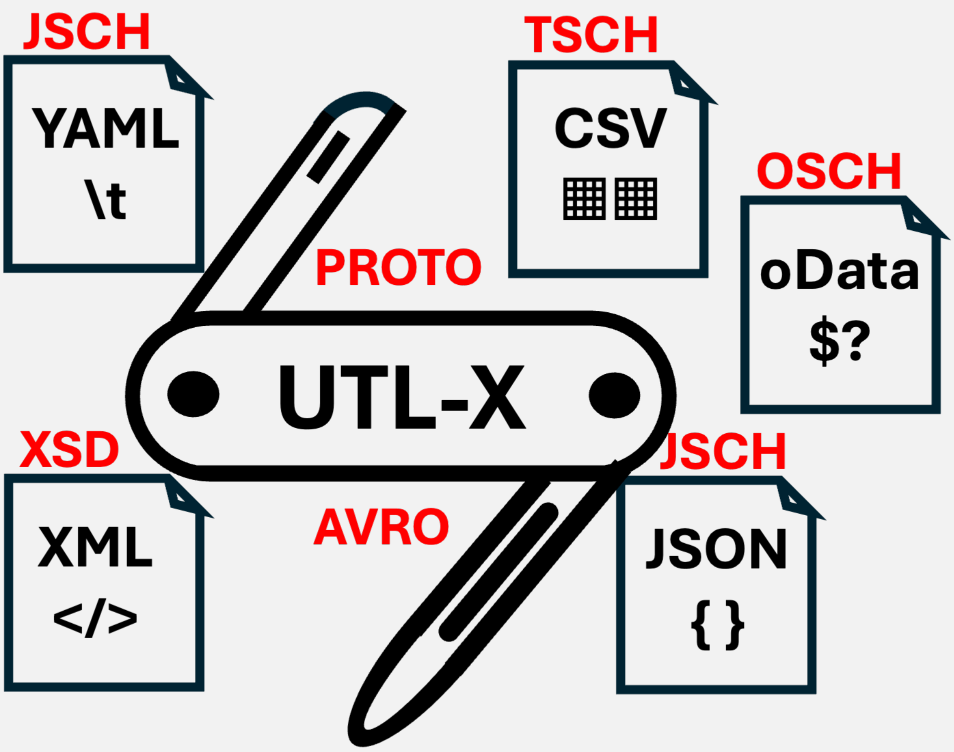


UTLXe

Production Transformation Engine



Deployment and Operations Guide

Running UTLXe on Azure Container Apps — From First Deploy to Production

Ir. Marcel A. Grauwen

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UTLXe on Azure — Deployment and Operations Guide

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UTL-X is open source software. The language specification, CLI tool, and standard library are freely available at `https://github.com/grauwen/utl-x`.

This is a companion guide to *UTL-X: One Language, All Formats* (ISBN 978-90-819728-1-9), which covers the complete language specification, standard library, and format system.

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Part I

Getting Started

Deploy, write your first transformation, and learn the Admin API

1 Quick Start

This chapter gets you from zero to a running transformation in under fifteen minutes. You will deploy UTLXe from the Azure Marketplace, upload a transformation, test it, and send your first real message.

1.1 What You Get

When you deploy UTLXe from the Azure Marketplace, Azure creates a Container App running the UTLXe production engine. The container exposes two ports:

- **Port 8085** — the data plane. Your applications send messages here for transformation. This port is exposed via the Container App ingress.
- **Port 8081** — the admin and health port. You manage transformations here. This port stays internal to your VNet.

The container starts empty — no transformations are loaded. You deploy them via the Admin API.

1.2 Deploy from the Azure Marketplace

1. Open the Azure Portal and search for "UTLXe" in the Marketplace.
2. Select your plan:
   - **Starter** (2 GB) — suitable for messages up to 50 KB, development and light production workloads.
   - **Professional** (4 GB) — suitable for messages up to 200 KB, production workloads with higher throughput.
3. Configure the deployment:
   - Resource group and region.
   - Container App name.
   - Persistent storage toggle — enable this if you want transformations to survive container restarts.
4. Click **Create**. The deployment takes approximately two minutes.

Once complete, note the Container App's internal IP (for the admin port) and the ingress URL (for the data plane).

1.3 Set the Admin Key

Before you can manage transformations, set the admin API key. This key protects the management endpoints from unauthorized access.

```
az containerapp secret set \
  -n utlxe -g myResourceGroup \
  --secrets admin-key=my-secret-key-here

az containerapp update \
  -n utlxe -g myResourceGroup \
  --set-env-vars UTLXE_ADMIN_KEY=secretref:admin-key
```

1.4 Verify the Deployment

Check that the container is running:

```
curl -s http://<internal-ip>:8081/health
```

```
{"status": "UP", "transformations": 0, "ready": false}
```

The engine is alive but has no transformations yet. The `ready: false` flag tells Kubernetes not to route traffic to this container.

1.5 Write Your First Transformation

Create a file called `hello.utlx`:

```
%utlx 1.0
input json
output json
---
{
  greeting: concat("Hello, ", $input.name, "!"),
  timestamp: now()
}
```

This transformation takes a JSON object with a `name` field and produces a greeting with a timestamp.

1.6 Upload It

```
curl -X POST \
  -H "X-Admin-Key: my-secret-key-here" \
  -F "source=@hello.utlx" \
  http://<internal-ip>:8081/admin/transformations/hello
```

```
{
  "status": "deployed",
  "name": "hello",
  "strategy": "COMPILED",
  "config": "defaults",
  "compiled_in_ms": 48
}
```

The transformation was compiled in 48 milliseconds and is ready to process messages. Check the health endpoint again:

```
{"status": "UP", "transformations": 1, "ready": true}
```

Now `ready: true` — the container will accept traffic.

1.7 Test It

Before routing real traffic, verify the transformation works with sample input:

```
curl -X POST \
  -H "X-Admin-Key: my-secret-key-here" \
  -H "Content-Type: application/json" \
  -d '{"name": "World"}' \
  http://<internal-ip>:8081/admin/transformations/hello/test
```

```
{
  "status": "ok",
  "output": {
    "greeting": "Hello, World!",
    "timestamp": "2026-05-05T14:30:00Z"
  },
```

```
  "duration_ms": 2
}
```

The test endpoint executes the transformation but does not count the call in Prometheus metrics. It is a safe way to verify before going live.

1.8 Send a Real Message

Now send a message through the data plane:

```
curl -X POST \
  -H "Content-Type: application/json" \
  -d '{"name": "Azure"}' \
  http://<ingress-url>:8085/api/transform/hello
```

```
{
  "greeting": "Hello, Azure!",
  "timestamp": "2026-05-05T14:30:05Z"
}
```

That is it. The transformation was compiled once at upload time. Every subsequent message executes the compiled version — typically one to five milliseconds per message.

1.9 What Just Happened

The following sequence describes the complete flow:

1. You uploaded `hello.utlx` to the Admin API on port 8081.
2. UTLXe parsed and compiled the transformation into an optimized in-memory representation.
3. The compiled transformation was registered in the engine and written to `/utlxe/data/transformations/hello/`.
4. The health endpoint switched to `ready: true`.
5. Your client sent a JSON message to the data plane on port 8085.
6. UTLXe looked up the compiled `hello` transformation, executed it, and returned the result.

The Admin API and the data plane run simultaneously — there is no "deployment mode" or downtime window.

1.10 Next Steps

- **Chapter 2** explains how to write more complex transformations — conditionals, array operations, multi-format output.
- **Chapter 3** covers the full Admin API — bundles, schemas, testing, and operational management.
- **Chapter 4** shows how to connect UTLXe to Azure Service Bus and Event Hub via Dapr.

2 Writing Transformations

This chapter teaches enough UTL-X to write production transformations. It is not the complete language reference — for that, see the companion book *UTL-X: One Language, All Formats*. Here you learn the patterns that cover ninety percent of real-world integration scenarios.

2.1 The .utlx File Structure

Every transformation has two parts: a header that declares the input and output formats, and a body that defines the transformation logic. They are separated by `---`.

```
%utlx 1.0
input json
output json
---
{
  orderId: $input.id,
  customer: $input.buyer.name,
  total: $input.amount
}
```

The header tells UTLXe how to parse the incoming message and how to serialize the output. The body is an expression that produces the output from the input.

2.2 Property Access

Access input fields with dot notation. The special variable `$input` refers to the incoming message.

```
$input.customer.name          // nested property
$input.items[0].price         // array index (0-based)
$input.@id                    // XML attribute
$input.order.item[2].@qty     // attribute on indexed element
```

If a property does not exist, the result is `null` — no error is thrown.

2.3 Object Construction

Curly braces create new objects. Each line is a key-value pair.

```
{
  orderId: $input.id,
  customer: $input.buyer.name,
  total: $input.quantity * $input.unitPrice
}
```

The spread operator copies all properties from an existing object:

```
{
  ...$input,
  processed: true,
  processedAt: now()
}
```

This produces the original input with two additional fields.

2.4 Let Bindings

Use `let` to name intermediate values. This avoids repetition and makes transformations readable.

```
{
  let subtotal = reduce($input.items, 0, (acc, i) -> acc + i.price)
  let tax = subtotal * 0.21
  let shipping = if (subtotal > 100) 0 else 9.95

  subtotal: subtotal,
  tax: round(tax, 2),
  shipping: shipping,
  total: round(subtotal + tax + shipping, 2)
}
```

2.5 Conditionals

The `if` expression chooses between two values:

```
if ($input.total > 1000) "premium" else "standard"
```

For multiple branches, use `match`:

```
match $input.status {
  "A" -> "Active",
  "I" -> "Inactive",
  "S" -> "Suspended",
  _ -> "Unknown"
}
```

The underscore _ is the catch-all pattern.

2.6 Array Operations

Arrays are transformed with higher-order functions. The three most common are `map`, `filter`, and `reduce`.

map — transform each element:

```
map($input.items, (item) -> {
  name: item.product,
  total: item.quantity * item.unitPrice
})
```

filter — keep elements matching a condition:

```
filter($input.items, (item) -> item.price > 100)
```

reduce — aggregate to a single value:

```
reduce($input.items, 0, (sum, item) -> sum + item.price)
```

These compose naturally:

```
$input.orders
  | filter(_, (o) -> o.status == "active")
  | map(_, (o) -> {id: o.id, total: o.amount})
  | sortBy(_, (o) -> o.total)
```

The pipe operator `|` passes the result of each step to the next. The underscore `_` is a placeholder for the piped value.

2.7 Common Standard Library Functions

UTL-X has over 650 standard library functions. The ones you will use most often:

2.7.1 Strings

`concat(a, b, c)`	Concatenate strings
`upperCase(s)` / `lowerCase(s)`	Case conversion
`trim(s)`	Remove leading and trailing whitespace
`replace(s, old, new)`	Replace substring
`contains(s, sub)`	Check if string contains substring
`substring(s, start, end)`	Extract substring
`length(s)`	String length
`split(s, delimiter)`	Split string into array
`startsWith(s, prefix)`	Check prefix

2.7.2 Arrays

`map(arr, fn)`	Transform each element
`filter(arr, fn)`	Keep matching elements
`reduce(arr, init, fn)`	Aggregate to single value
`find(arr, fn)`	First matching element
`sortBy(arr, fn)`	Sort by key
`groupBy(arr, fn)`	Group by key
`flatMap(arr, fn)`	Map and flatten
`size(arr)`	Array length
`first(arr)` / `last(arr)`	First or last element

2.7.3 Math and Type

`round(n, decimals)`	Round to decimal places
`floor(n)` / `ceil(n)`	Floor or ceiling
`abs(n)` / `min(a, b)` / `max(a, b)`	Absolute value, min, max
`toString(v)` / `toNumber(v)`	Type conversion
`isNull(v)`	Null check
`now()`	Current timestamp
`formatDate(d, pattern)`	Format a date

2.8 Multi-Format Transformations

The header declares formats. The body works the same regardless of whether the input is JSON, XML, CSV, or YAML — UTL-X operates on the Universal Data Model (UDM), which is format-agnostic.

JSON to XML:

```
%utlx 1.0
input json
output xml
---
{
  Invoice: {
    ID: $input.orderId,
    Amount: $input.total,
    Currency: $input.currency
  }
}
```

The XML serializer produces `<Invoice><ID>12345</ID><Amount>250.0</Amount>...</Invoice>` from this object. You write objects — the format system handles serialization.

XML to JSON:

```
%utlx 1.0
input xml
output json
---
{
  orderId: $input.Invoice.ID,
  total: toNumber($input.Invoice.Amount),
  currency: $input.Invoice.Currency
}
```

CSV to JSON:

```
%utlx 1.0
input csv {header: true}
output json
---
map($input, (row) -> {
  name: row.Name,
  email: lowerCase(row.Email),
  active: row.Status == "A"
})
```

2.9 User-Defined Functions

Define reusable logic with `function`:

```
function discount(amount, tier) {
  match tier {
    "gold" -> amount * 0.10,
    "silver" -> amount * 0.05,
    _ -> 0
  }
}

{
  let disc = discount($input.total, $input.customerTier)
  orderId: $input.id,
  subtotal: $input.total,
```

```
  discount: round(disc, 2),
  total: round($input.total - disc, 2)
}
```

Functions are defined before the main expression. They can call other functions and use all standard library functions.

2.10 Worked Example: Invoice Transformation

A complete transformation that converts a Dynamics 365 Business Central order into a simplified UBL-like invoice:

```
%utlx 1.0
input json
output xml
---
{
  let order = $input
  let subtotal = reduce(order.lines, 0,
    (sum, line) -> sum + line.quantity * line.unitPrice)
  let tax = subtotal * 0.21

  Invoice: {
    ID: concat("INV-", order.orderNumber),
    IssueDate: formatDate(now(), "yyyy-MM-dd"),
    CustomerName: order.customer.name,
    Currency: order.currency,
    InvoiceLines: {
      InvoiceLine: map(order.lines, (line) -> {
        ID: line.lineNumber,
        Quantity: line.quantity,
        UnitPrice: line.unitPrice,
        LineTotal: round(line.quantity * line.unitPrice, 2)
      })
    },
    Subtotal: round(subtotal, 2),
    Tax: round(tax, 2),
    Total: round(subtotal + tax, 2)
  }
}
```

The XML serializer produces the UBL structure from this object. You write objects with property names that become element names — the format system handles the rest.

This transformation demonstrates property access, array iteration with `map`, aggregation with `reduce`, string concatenation, date formatting, and arithmetic.

2.11 Where to Learn More

- The complete language specification, including pattern matching, recursive functions, and the full operator table, is in the companion book *UTL-X: One Language, All Formats*.
- The standard library reference (all 650+ functions with examples) is in chapters 16 and 17 of the companion book.
- Example transformations are available in the project repository under `examples/`.

3 The Admin API

The Admin API is the primary interface for deploying and operating UTLXe on Azure. It runs on port 8081 alongside the health and metrics endpoints. This chapter covers every operation you need for day-to-day management.

3.1 Architecture: Two Ports, Two Purposes

UTLXe exposes two HTTP servers on separate ports:

Port	Purpose	Access
8081	Admin API + health + metrics	Internal to VNet — not exposed via ingress
8085	Data plane (message processing)	Exposed via Container App ingress

This separation allows network isolation. The data plane is accessible to client applications and Dapr sidecars. The admin port is restricted to operators and CI/CD pipelines within the VNet.

Both ports are always active. There is no mode switch — you can upload transformations while the data plane processes messages. See the architecture decision in the design document for the rationale.

3.2 Authentication

Every request to `/admin/*` endpoints must include the `X-Admin-Key` header:

```
curl -H "X-Admin-Key: my-secret-key-here" \
  http://<internal-ip>:8081/admin/transformations
```

The key is set via the `UTLXE_ADMIN_KEY` environment variable. If this variable is not set, all admin endpoints return 403 — the API is locked by default.

The health endpoints (`/health`, `/metrics`) do not require authentication. They are read-only and needed by Kubernetes probes and Prometheus.

3.3 Uploading Transformations

3.3.1 Single .utlx File

The simplest deployment: upload just the `.utlx` source file. UTLXe applies sensible defaults (COMPILED strategy, auto-detect format).

```
curl -X POST \
  -H "X-Admin-Key: $KEY" \
  -F "source=@invoice-to-ubl.utlx" \
  http://<admin>:8081/admin/transformations/invoice-to-ubl
```

Response:

```
{
  "status": "deployed",
  "name": "invoice-to-ubl",
  "strategy": "COMPILED",
  "config": "defaults",
  "compiled_in_ms": 48
}
```

3.3.2 With Configuration

For explicit control over strategy and validation, include a `transform.yaml`:

```
curl -X POST \
  -H "X-Admin-Key: $KEY" \
  -F "source=@invoice-to-ubl.utlx" \
  -F "config=@transform.yaml" \
  http://<admin>:8081/admin/transformations/invoice-to-ubl
```

A typical `transform.yaml`:

```yaml
strategy: COMPILED
validationPolicy: strict
inputs:
  - name: input
    schema: order.json
maxConcurrent: 4
```

3.3.3 ZIP Bundle

For batch deployment, upload a ZIP containing all transformations and schemas at once. This replaces the entire bundle atomically.

```
curl -X POST \
  -H "X-Admin-Key: $KEY" \
  -F "file=@bundle.zip" \
  http://<admin>:8081/admin/bundle
```

The ZIP follows this structure:

```
bundle.zip
  schemas/
    order.xsd
    invoice.json
  transformations/
    invoice-to-ubl/
      invoice-to-ubl.utlx
    order-enrichment/
      order-enrichment.utlx
```

The `schemas/` directory is optional. Each transformation directory requires only the `.utlx` source file. A `transform.yaml` can be added alongside the `.utlx` to override defaults (strategy, validation policy, output binding) — but it is not required.

3.4 Managing Schemas

Schemas are shared resources, stored separately from transformations. A single schema like `order.xsd` can be referenced by multiple transformations.

Upload a schema:

```
curl -X POST \
  -H "X-Admin-Key: $KEY" \
  -F "file=@order.xsd" \
  http://<admin>:8081/admin/schemas/order.xsd
```

List all schemas:

```
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/schemas
```

```
{
  "schemas": [
    {"filename": "order.xsd", "size_bytes": 4820, "uploaded_at": "2026-05-05T14:29:00Z"},
    {"filename": "invoice.json", "size_bytes": 2340, "uploaded_at":
"2026-05-05T14:29:05Z"}
  ]
}
```

Updating a schema does not recompile transformations — schemas are resolved at validation time, not compile time. The new schema takes effect on the next message.

3.5 Testing Before Go-Live

The test endpoint is the most important step after uploading a transformation. It executes the transformation with sample input and returns the result without affecting metrics.

```
curl -X POST \
  -H "X-Admin-Key: $KEY" \
  -H "Content-Type: application/json" \
  -d '{"orderId": "12345", "amount": 250.00, "currency": "EUR"}' \
  http://<admin>:8081/admin/transformations/invoice-to-ubl/test
```

On success:

```
{
  "status": "ok",
  "output": {"Invoice": {"ID": "INV-12345", "Amount": 250.0, "Currency": "EUR"}},
  "duration_ms": 3
}
```

On failure:

```
{
  "status": "error",
  "error": "Null reference: $input.customer.address",
  "line": 14,
  "column": 22
}
```

Always test before routing real traffic. The deployment workflow should be: upload, test, then allow traffic.

3.6 Listing and Inspecting

List all deployed transformations:

```
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/transformations
```

```json
{
  "transformations": [
    {
      "name": "invoice-to-ubl",
      "strategy": "COMPILED",
      "status": "ready",
      "config": "explicit",
      "deployed_at": "2026-05-05T14:30:00Z",
      "messages_processed": 12345,
      "avg_transform_ms": 2.3
    }
  ]
}
```

Get details for a specific transformation, including the source code:

```
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/transformations/invoice-to-ubl
```

3.7 Updating Transformations

Upload the same name again to replace a transformation. The update is atomic:

1. The new source is compiled.
2. If compilation fails, the upload is rejected — the running transformation is untouched.
3. If compilation succeeds, the new version replaces the old one in the registry.
4. In-flight messages on the old version complete normally.
5. New messages use the new version immediately.

There is no downtime, no mode switch, and no restart.

3.8 Deleting

Remove a single transformation:

```
curl -X DELETE \
  -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/transformations/invoice-to-ubl
```

Remove everything and start fresh:

```
curl -X DELETE \
  -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/bundle
```

3.9 Exporting the Bundle

Download the current state as a ZIP file:

```
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/bundle -o bundle-backup.zip
```

This is useful for version control, backup, or migrating transformations to another container. The exported ZIP has the same format as the upload — you can re-import it with `POST /admin/bundle`.

3.10 Data Plane Discovery

Client applications can discover available transformations without an admin key. This endpoint runs on the data plane port (8085):

```
curl http://<ingress>:8085/api/transformations
```


```
{
  "transformations": [
    {"name": "invoice-to-ubl", "status": "ready", "input": "json", "output": "xml"},
    {"name": "order-enrichment", "status": "ready", "input": "json", "output": "json"}
  ]
}
```


This makes the data plane self-describing. A client connecting for the first time can discover what transformations are available without out-of-band knowledge.

3.11 Engine Info

Get basic operational information:

```
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/info
```


```
{
  "version": "1.0.1",
  "uptime_seconds": 86400,
  "mode": "http",
  "workers": 4,
  "heap_max_mb": 1536,
  "data_dir": "/utlxe/data",
  "persistence": "volume-backed",
  "admin_key_set": true,
  "transformations": 3,
  "schemas": 2,
  "ready": true
}
```


3.12 Two Deployment Workflows

The Admin API supports two workflows. Use whichever fits your process.

Batch workflow — assemble a ZIP in your CI/CD pipeline and deploy in one call:

```
curl -X POST -H "X-Admin-Key: $KEY" \
  -F "file=@bundle.zip" \
  http://<admin>:8081/admin/bundle
```

Incremental workflow — upload resources one at a time:

```
# Upload schemas first
curl -X POST -H "X-Admin-Key: $KEY" -F "file=@order.xsd" .../admin/schemas/order.xsd

# Then transformations
```

```
curl -X POST -H "X-Admin-Key: $KEY" -F "source=@invoice.utlx" .../admin/transformations/
invoice

# Export the assembled bundle for version control
curl -H "X-Admin-Key: $KEY" .../admin/bundle -o my-bundle.zip
```

Both workflows produce the same result. You can start with the incremental workflow during development, then switch to batch for production CI/CD.

Part II

Integration

Connect to Azure services and monitor your deployment

4 Connecting to Azure Services

UTLXe integrates with Azure messaging services through a Dapr sidecar. Dapr handles the connection to Service Bus, Event Hub, and other Azure services, while UTLXe focuses on the transformation logic. This chapter covers the most common integration patterns.

4.1 How Dapr Works with UTLXe

Dapr runs as a sidecar container alongside UTLXe in the same Container App. It acts as a message broker adapter:

1. A message arrives on Azure Service Bus (or Event Hub).
2. Dapr receives the message and forwards it to UTLXe via HTTP: `POST /api/dapr/input/{bindingName}`.
3. UTLXe transforms the message.
4. UTLXe sends the result back to Dapr: `POST http://localhost:3500/v1.0/bindings/{outputBinding}`.
5. Dapr delivers the transformed message to the output queue or topic.
6. UTLXe acknowledges the original message (HTTP 200), and Dapr completes it on Service Bus.

UTLXe never connects directly to Service Bus or Event Hub. Dapr abstracts the messaging infrastructure, which means the same UTLXe container works with different messaging systems by changing only the Dapr configuration.

4.2 Azure Service Bus

4.2.1 Receiving from a Queue

Configure a Dapr input binding that delivers messages to UTLXe. The binding name maps to the transformation name.

Dapr component (`servicebus-input.yaml`):

```yaml
apiVersion: dapr.io/v1alpha1
kind: Component
metadata:
  name: orders-in
spec:
  type: bindings.azure.servicebusqueues
  metadata:
    - name: connectionString
      secretKeyRef:
        name: servicebus-connection
        key: connectionString
    - name: queueName
      value: "incoming-orders"
    - name: maxConcurrentHandlers
      value: "10"
```

When a message arrives on the `incoming-orders` queue, Dapr calls `POST /api/dapr/input/orders-in` on UTLXe. UTLXe looks up a transformation named `orders-in` (or uses the `X-UTLXe-Transform` header to specify a different one) and processes the message.

4.2.2 Sending to a Queue

Configure a Dapr output binding for the transformed messages:

```yaml
apiVersion: dapr.io/v1alpha1
kind: Component
metadata:
  name: orders-out
```

```yaml
spec:
  type: bindings.azure.servicebusqueues
  metadata:
    - name: connectionString
      secretKeyRef:
        name: servicebus-connection
        key: connectionString
    - name: queueName
      value: "processed-orders"
```

In the transformation's `transform.yaml`, specify the output binding:

```yaml
strategy: COMPILED
outputBinding: orders-out
```

After transformation, UTLXe automatically sends the result to Dapr's output binding, which delivers it to the `processed-orders` queue.

4.2.3 Topics and Subscriptions

For publish/subscribe patterns, use Service Bus topics instead of queues. The Dapr component type changes to `bindings.azure.servicebustopics`, but the UTLXe integration is identical — Dapr delivers messages the same way.

4.2.4 Dead-Letter Queue

When a transformation fails, UTLXe returns HTTP 500 to Dapr. Dapr abandons the message on Service Bus, which increments its delivery count. After the maximum number of retries (configured on the Service Bus queue), the message moves to the dead-letter queue.

You can inspect recent errors via the Admin API:

```bash
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/transformations/orders-in/errors
```

This returns the last 100 errors with timestamps, error messages, and a preview of the input that caused the failure.

4.3 Azure Event Hub

Event Hub integration follows the same pattern as Service Bus, with a different Dapr component type.

Input binding (`eventhub-input.yaml`):

```yaml
apiVersion: dapr.io/v1alpha1
kind: Component
metadata:
  name: events-in
spec:
  type: bindings.azure.eventhubs
  metadata:
    - name: connectionString
      secretKeyRef:
        name: eventhub-connection
        key: connectionString
    - name: consumerGroup
      value: "utlxe-consumer"
```

```yaml
    - name: storageAccountName
      value: "stcheckpoints"
    - name: storageContainerName
      value: "checkpoints"
```

Event Hub uses consumer groups and checkpointing for partition management. Dapr handles checkpointing automatically — UTLXe does not need to manage offsets.

Output binding:

```yaml
apiVersion: dapr.io/v1alpha1
kind: Component
metadata:
  name: events-out
spec:
  type: bindings.azure.eventhubs
  metadata:
    - name: connectionString
      secretKeyRef:
        name: eventhub-connection
        key: connectionString
```

4.4 Direct HTTP (Without Dapr)

Not all integrations require Dapr. For simple request/response patterns, clients can call the data plane directly:

```bash
curl -X POST \
  -H "Content-Type: application/json" \
  -d @order.json \
  http://<ingress>:8085/api/transform/invoice-to-ubl
```

This is suitable for:

- API gateway integration — transform requests or responses inline.
- Batch processing — send messages from a script or pipeline.
- Testing and development — call the transformation directly.

4.5 Worked Example: Dynamics 365 to Peppol

A complete end-to-end flow for the European e-invoicing scenario:

1. **Dynamics 365 Business Central** publishes an invoice as OData JSON to a Service Bus queue.
2. **Dapr** delivers the message to UTLXe.
3. **UTLXe** runs three chained transformations:
   - `invoice-to-ubl` — converts D365 JSON to UBL 2.1 XML.
   - `validate-ubl` — validates the UBL against Peppol BIS 3.0 rules.
   - `route-by-country` — adds country-specific tax rules.
4. **UTLXe** sends the final UBL XML to Dapr's output binding.
5. **Dapr** publishes to an output Service Bus topic.
6. A **Peppol Access Point** consumes from the topic and delivers the invoice via AS4.

All three transformations are uploaded as a single bundle:

```
curl -X POST -H "X-Admin-Key: $KEY" \
  -F "file=@peppol-bundle.zip" \
  http://<admin>:8081/admin/bundle
```

The bundle contains the three `.utlx` files, their `transform.yaml` configs specifying the pipeline order, and the UBL and Peppol schemas for validation.

5 Monitoring and Observability

Production deployments need visibility into what UTLXe is doing. This chapter covers health probes, Prometheus metrics, and how to set up monitoring dashboards.

5.1 Health Endpoint

The health endpoint runs on port 8081 and serves two purposes: Kubernetes probes and operational status checks.

```
curl http://<internal-ip>:8081/health
```

```
{
  "status": "UP",
  "transformations": 3,
  "ready": true
}
```

The fields:

`status`	`"UP"` if the process is alive. Used by the liveness probe.
`transformations`	Number of compiled transformations in the registry.
`ready`	`true` when at least one transformation is loaded and compiled. Used by the readiness probe.

5.1.1 Liveness vs. Readiness

Probe	Checks	Action if fails
Liveness	`status == "UP"`	Kubernetes restarts the container
Readiness	`ready == true`	Kubernetes stops routing traffic to this instance

A container that just started but has not received its bundle yet is **alive but not ready**. Traffic is held until transformations are uploaded and compiled. This prevents errors during the deployment window.

5.2 Prometheus Metrics

UTLXe exposes Prometheus metrics on the same port:

```
curl http://<internal-ip>:8081/metrics
```

5.2.1 Per-Transformation Metrics

```
utlxe_messages_total{transform="invoice-to-ubl"} 12345
utlxe_transform_duration_seconds{transform="invoice-to-ubl",quantile="0.50"} 0.002
utlxe_transform_duration_seconds{transform="invoice-to-ubl",quantile="0.95"} 0.005
utlxe_transform_duration_seconds{transform="invoice-to-ubl",quantile="0.99"} 0.012
utlxe_errors_total{transform="invoice-to-ubl"} 3
```

5.2.2 Engine-Level Metrics

```
utlxe_active_threads 4
utlxe_heap_used_bytes 524288000
utlxe_uptime_seconds 86400
```

These metrics are standard Prometheus format and can be scraped by any Prometheus-compatible system.

5.3 Setting Up Azure Monitor

Azure Container Apps integrates with Azure Monitor out of the box. Container logs are available in the Log Analytics workspace associated with the Container App Environment.

To view container logs:

```
az containerapp logs show \
  -n utlxe -g myResourceGroup \
  --follow
```

UTLXe logs in structured format: timestamp, level, transformation name, and message. Filter by transformation name to isolate issues.

For Prometheus metrics, configure Azure Monitor managed Prometheus or deploy a Prometheus instance that scrapes port 8081.

5.4 Grafana Dashboard

A recommended Grafana dashboard layout for UTLXe:

Row 1 — Throughput:
- Messages per second (rate of `utlxe_messages_total`)
- Errors per second (rate of `utlxe_errors_total`)
- Error rate percentage

Row 2 — Latency:
- p50 latency (median transformation time)
- p95 latency (tail latency)
- p99 latency (worst case)

Row 3 — Resources:
- Heap usage (percentage of max)
- Active threads
- Container CPU and memory from Azure Monitor

5.4.1 Alert Rules

Configure alerts for:

Metric	Threshold	Action
Error rate	> 1% for 5 minutes	Notify on-call
p99 latency	> 500ms for 5 minutes	Investigate — possible GC pressure or large messages
Heap usage	> 80% of max	Warning — consider upgrading plan
Transformations	**5.5 0 for 2 minutes**	Container restarted without bundle — check persistence

5.6 Error Diagnosis

Prometheus gives you counters, but not the actual error messages. For quick diagnosis, use the error ring buffer:

```
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/transformations/invoice-to-ubl/errors?limit=10
```


```
{
  "errors": [
    {
      "timestamp": "2026-05-05T14:32:01Z",
      "message": "Null reference: $input.customer.address",
      "line": 14,
      "input_preview": "{\"orderId\":\"12345\",\"customer\":{\"name\":\"Acme\"}}"
    }
  ],
  "total_errors": 47,
  "showing": 10
}
```


The `input_preview` is truncated to 200 characters to avoid exposing full message payloads. This is usually enough to identify the pattern that causes failures.

The typical diagnosis workflow:

1. Notice elevated error rate in Grafana.
2. Check the error ring buffer for the affected transformation.
3. Identify the pattern (missing field, unexpected type, schema mismatch).
4. Pause the transformation if needed (Chapter 6).
5. Fix and re-upload.
6. Test with sample input.
7. Resume.

Part III

Production

Operations, scaling, security, and automation

6 Operations

This chapter covers day-to-day operational tasks: updating transformations without downtime, handling incidents, and managing validation at runtime.

6.1 Updating a Transformation

Upload a new version of the same transformation name. The update is atomic and zero-downtime:

```
curl -X POST \
  -H "X-Admin-Key: $KEY" \
  -F "source=@invoice-to-ubl-v2.utlx" \
  http://<admin>:8081/admin/transformations/invoice-to-ubl
```

What happens internally:

1. UTLXe compiles the new source. If compilation fails, the upload is rejected and the running version is untouched.
2. The new compiled version atomically replaces the old one in the registry.
3. Messages that are currently being processed on the old version complete normally — they hold a reference to the old compiled code.
4. New messages arriving after the swap use the new version.

There is no restart, no mode switch, and no window where messages are dropped.

6.2 Pause and Resume

During an incident, you may need to stop processing for a specific transformation without removing it. Pausing preserves the compiled transformation and its on-disk state while stopping traffic.

Pause:

```
curl -X POST \
  -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/transformations/invoice-to-ubl/pause
```

While paused:

- The data plane returns 503 for that transformation. Other transformations continue normally.
- Dapr receives the 503 and does not acknowledge the message. Service Bus retries it later (or moves it to the dead-letter queue after max retries).
- The transformation stays compiled in memory and on disk — resume is instant.
- The listing shows `"status": "paused"`.

Resume:

```
curl -X POST \
  -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/transformations/invoice-to-ubl/resume
```

Processing resumes immediately. Queued messages on Service Bus are delivered again.

6.3 Incident Response Workflow

A complete incident lifecycle:

1. **Detect** — Grafana alert fires on elevated error rate.
2. **Diagnose** — Check the error ring buffer:

```
curl -H "X-Admin-Key: $KEY" \
  .../admin/transformations/invoice-to-ubl/errors
```

3. **Contain** — Pause the transformation:

```
curl -X POST -H "X-Admin-Key: $KEY" \
  .../admin/transformations/invoice-to-ubl/pause
```

4. **Fix** — Edit the `.utlx` source to handle the failing case.
5. **Deploy** — Upload the fix:

```
curl -X POST -H "X-Admin-Key: $KEY" \
  -F "source=@invoice-to-ubl-fixed.utlx" \
  .../admin/transformations/invoice-to-ubl
```

6. **Verify** — Test with the input that caused the failure:

```
curl -X POST -H "X-Admin-Key: $KEY" \
  -d '{"orderId":"12345","customer":null}' \
  .../admin/transformations/invoice-to-ubl/test
```

7. **Resume** — Bring the transformation back online:

```
curl -X POST -H "X-Admin-Key: $KEY" \
  .../admin/transformations/invoice-to-ubl/resume
```

The entire cycle — from detection to resume — can happen in minutes without a container restart.

6.4 Validation Override

Schema validation catches malformed messages before they reach the transformation logic. But during an incident, validation might be too strict — blocking messages that are technically valid but don't match the expected schema exactly.

Set a runtime override:

```
curl -X POST \
  -H "X-Admin-Key: $KEY" \
  -H "Content-Type: application/json" \
  -d '{"policy": "off"}' \
  http://<admin>:8081/admin/transformations/invoice-to-ubl/validation
```

Check the effective state:

```
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/transformations/invoice-to-ubl/validation
```

```
{
  "effective_policy": "off",
  "source": "runtime-override",
  "config_policy": "strict",
  "header_policy": "strict"
}
```

Remove the override to revert to the configured policy:

```
curl -X DELETE \
  -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/transformations/invoice-to-ubl/validation
```

The override is ephemeral — it is not written to disk and disappears on container restart. The precedence chain is:

Level	Set by	Persists
Runtime override	Ops via Admin API	No — ephemeral
`transform.yaml` config	Ops at deploy time	Yes — on disk
`.utlx` header	Developer at dev time	Yes — in source
Default	—	`strict`

The highest-priority level that is set wins.

6.5 Updating Schemas

Replace a schema without recompiling transformations:

```
curl -X POST \
  -H "X-Admin-Key: $KEY" \
  -F "file=@order-v2.xsd" \
  http://<admin>:8081/admin/schemas/order.xsd
```

The new schema takes effect on the next message. Transformations that reference `order.xsd` do not need to be re-uploaded — schema resolution happens at validation time, not compile time.

6.6 Bulk Operations

Replace everything at once (atomic):

```
curl -X POST -H "X-Admin-Key: $KEY" \
  -F "file=@new-bundle.zip" \
  http://<admin>:8081/admin/bundle
```

Remove everything and start fresh:

```
curl -X DELETE -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/bundle
```

Export the current state as a backup:

```
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/bundle -o backup.zip
```

6.7 Engine Configuration

View the current engine configuration:

```
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/config
```

Update a runtime-safe field:

```
curl -X POST \
  -H "X-Admin-Key: $KEY" \
  -H "Content-Type: application/json" \
  -d '{"maxInputSize": "10MB"}' \
  http://<admin>:8081/admin/config
```

Fields that require a restart (like port numbers) are accepted but flagged in the response:

```
{"updated": ["maxInputSize"], "restart_required": []}
```

7 Persistence and Scaling

This chapter explains how transformations survive container restarts and how to scale UTLXe for production throughput.

7.1 The Persistence Problem

Docker containers have an ephemeral filesystem. When a container restarts — due to a crash, a scale-to-zero event, or a redeployment — everything written inside the container is lost. This means the `/utlxe/data/` directory, where uploaded transformations and schemas are stored, is wiped clean.

This is a fundamental container property, not something UTLXe can fix internally. The solution is external storage.

7.2 Azure Files Volume Mount

The recommended approach for the Azure Marketplace is an Azure File Share mounted at `/utlxe/data/`. The mount is configured by the platform at the infrastructure level — before the container starts. UTLXe uses standard Java file I/O to read and write the directory. It has no knowledge that the directory is backed by a network file share.

```
Container filesystem (ephemeral):
  /utlxe/utlxe.jar             from Docker image, rebuilt on restart

Mount point (persistent):
  /utlxe/data/                  Azure File Share — survives restarts
    schemas/
      order.xsd
    transformations/
      invoice-to-ubl/
        invoice-to-ubl.utlx
      order-enrichment/
        order-enrichment.utlx
```

On restart, UTLXe scans `/utlxe/data/`, finds the transformations from the previous session, compiles them, and becomes ready. Zero manual intervention.

To enable persistent storage, check the "Enable persistent transformation storage" option during Marketplace deployment. This creates an Azure Storage Account and File Share, and configures the volume mount in the Container App.

7.3 CI/CD Re-Deploy Pattern

An alternative to persistent storage: let the CI/CD pipeline re-deploy transformations after every container start.

1. Container starts with an empty `/utlxe/data/`.
2. Health endpoint returns `ready: false`.
3. The CI/CD pipeline detects the not-ready state and uploads the bundle.
4. UTLXe compiles the transformations.
5. Health endpoint switches to `ready: true`.
6. Kubernetes routes traffic.

The bundle lives in the CI/CD system (a git repository or artifact store). The container is truly stateless — the pipeline is the source of truth.

7.4 Startup Sequence

Regardless of persistence mode, the startup sequence is:

1. Start the HTTP server on port 8081 (health, metrics, admin API).
2. Scan `/utlxe/data/` for existing transformations and schemas.
   - If found (volume mount from previous session): compile and register them.
   - If empty: wait for API uploads.
3. Start the data plane on port 8085.
4. The readiness probe checks `ready == true` before Kubernetes routes traffic.

The admin API is available from step 1 — you can upload transformations while the startup scan is still running.

7.5 Memory Sizing

UTLXe runs on the JVM with a configured heap size. The heap must be large enough to hold the message being processed plus the intermediate objects created during transformation.

Plan	Container	Heap	Max message
Starter	2 GB	1536 MB	50 KB
Professional	4 GB	3072 MB	200 KB

The heap is set to 75% of the container memory. The remaining 25% is for the JVM itself (metaspace, thread stacks, native memory) and the operating system.

The `-XX:+AlwaysPreTouch` JVM flag allocates the entire heap at startup. If the container does not have enough RAM, the process fails immediately with a clear error — rather than crashing hours later under load.

7.6 Horizontal Scaling

For higher throughput, deploy multiple container replicas behind the Azure load balancer. Each replica runs an independent UTLXe instance.

If using persistent storage, all replicas share the same Azure File Share. Upload transformations once — all replicas see the same files.

If using CI/CD re-deploy, the pipeline must deploy to each replica (or use a shared storage backend for the bundle).

Azure Service Bus distributes messages across consumers automatically. Event Hub uses consumer groups to distribute partitions across replicas.

7.7 Never Use Swap

Swap and garbage collection do not mix. When the JVM's garbage collector scans the heap, it touches every page. If those pages have been swapped to disk, each page access triggers a page fault — a disk I/O operation that takes milliseconds instead of nanoseconds. A GC pause that normally takes 20 milliseconds can take 20 *seconds* when pages are swapped.

The rule: **never run UTLXe with swap enabled.**

In Azure Container Apps, set the memory request equal to the memory limit:

```yaml
resources:
  requests:
    memory: "2Gi"
  limits:
    memory: "2Gi"
```

This ensures the container gets exactly the memory it requests, with no swap. If the container exceeds its memory limit, Kubernetes kills it (OOM) — which is faster to recover from than swap thrashing.

7.8 Poison Messages

A poison message is an input that is too large or too malformed to process. Without protection, a single 2 GB message can exhaust the JVM heap and crash the container.

The `maxInputSize` configuration rejects oversized messages before parsing:

```
maxInputSize: 5MB
```

Messages larger than this limit are rejected immediately with a 413 status code. The Dapr sidecar receives the error, and Service Bus moves the message to the dead-letter queue after max retries.

The default is 5 MB. Increase it only if you know your messages are legitimately larger, and verify that the heap has enough room.

8 Security

This chapter covers the security model of a UTLXe deployment: network isolation, authentication, secrets management, and the principle of least privilege.

8.1 Network Architecture

UTLXe uses port separation to isolate management traffic from data traffic:

Port	Scope	Contains
8085	Public (via ingress)	Data plane — message processing only
8081	Internal (VNet only)	Admin API, health probes, Prometheus metrics

The Container App ingress exposes only port 8085 to the outside world. Port 8081 is accessible only within the VNet, which means:

- Client applications can send messages for transformation but cannot upload, delete, or modify transformations.
- Operators and CI/CD pipelines access the admin API from within the VNet (or via VPN/private endpoint).
- Prometheus scrapes metrics from port 8081 inside the VNet.
- Kubernetes probes check health on port 8081 inside the VNet.

8.2 Admin API Authentication

The admin API is protected by an API key. Every request to `/admin/*` must include:

```
X-Admin-Key: <value of UTLXE_ADMIN_KEY>
```

If `UTLXE_ADMIN_KEY` is not set, all admin endpoints return 403. The API is locked by default — there is no open-by-default behavior.

Store the key as a Container App secret:

```
az containerapp secret set \
  -n utlxe -g myResourceGroup \
  --secrets admin-key=<your-secret-key>
```

Reference it in the environment:

```
az containerapp update \
  -n utlxe -g myResourceGroup \
  --set-env-vars UTLXE_ADMIN_KEY=secretref:admin-key
```

Container App secrets are encrypted at rest and injected as environment variables at runtime. They are not visible in logs or in the container filesystem.

To rotate the key, update the secret and restart the container. No code change is needed.

8.3 Non-Root Container

The UTLXe Docker image runs as a non-root user (`utlxe:utlxe`). The Dockerfile creates a dedicated user and group:

```
RUN groupadd -r utlxe && useradd -r -g utlxe utlxe
USER utlxe
```

The container has no elevated privileges. The only writable directory is `/utlxe/data/`, where the Admin API stores uploaded transformations and schemas.

8.4 Secrets in Transformations

Transformations sometimes need secrets — API keys, connection strings, tokens. Never hardcode secrets in `.utlx` files. Instead, use the `env()` function to read environment variables:

```
%utlx 1.0
input json
output json
---
{
  ...$input,
  authHeader: concat("Bearer ", env("API_TOKEN"))
}
```

The `API_TOKEN` value comes from a Container App secret, injected as an environment variable. The `.utlx` source can be committed to version control without exposing the secret.

The `env()` function is restricted in the CLI for security, but available in the engine (UTLXe) where environment variables are controlled by the deployment configuration.

8.5 VNet Integration

For production deployments, place the Container App Environment in a VNet:

- The admin port (8081) is accessible only from within the VNet.
- The data plane ingress (8085) can be configured as internal (VNet only) or external (public with TLS).
- Service Bus and Storage Account connections use private endpoints — no traffic over the public internet.

A typical production network layout:

UTLXe Container App	VNet, private subnet
Admin access	VNet only (or via Azure Bastion / VPN)
Data plane ingress	Internal or external, TLS terminated by Azure
Service Bus	Private endpoint in same VNet
Storage Account	Private endpoint in same VNet (for Azure Files)
Prometheus	In same VNet, scrapes port 8081

8.6 TLS and HTTPS

UTLXe itself runs HTTP on both ports. TLS termination is handled by the Container App ingress for port 8085. Internal traffic between Dapr, UTLXe, and other services within the VNet does not need TLS — it is already isolated by the network boundary.

If your security policy requires TLS on internal traffic, configure the Container App Environment with mTLS (mutual TLS), which encrypts all traffic between containers in the environment.

9 CI/CD Integration

Automated deployment ensures that transformations are tested, versioned, and deployed consistently. This chapter shows how to integrate UTLXe with GitHub Actions and Azure DevOps.

9.1 The Deployment Model

The transformation source lives in a git repository. The CI/CD pipeline:

1. Assembles a bundle (ZIP) from the repository contents.
2. Validates the bundle against the running UTLXe instance (dry run).
3. Deploys the bundle.
4. Verifies with test inputs.

No custom Docker image is built. The UTLXe container from the Marketplace stays as-is. Only the transformations change.

9.2 Repository Structure

A recommended layout for the transformation repository:

```
my-transformations/
  schemas/
    order.xsd
    invoice.json
  transformations/
    invoice-to-ubl/
      invoice-to-ubl.utlx
    order-enrichment/
      order-enrichment.utlx
  test-data/
    invoice-to-ubl/
      input.json
      expected-output.xml
  scripts/
    build-bundle.sh
    deploy.sh
```

The `test-data/` directory contains sample inputs and expected outputs for each transformation. The pipeline uses these for post-deployment verification.

9.3 Building the Bundle

A simple script assembles the ZIP:

```bash
#!/bin/bash
# build-bundle.sh
cd "$(dirname "$0")/.."
zip -r bundle.zip schemas/ transformations/
echo "Bundle created: bundle.zip"
```

9.4 GitHub Actions Workflow

```yaml
name: Deploy Transformations
on:
  push:
    branches: [main]
```

```yaml
env:
  UTLXE_ADMIN_URL: ${{ secrets.UTLXE_ADMIN_URL }}
  UTLXE_ADMIN_KEY: ${{ secrets.UTLXE_ADMIN_KEY }}
  UTLXE_DATA_URL: ${{ secrets.UTLXE_DATA_URL }}

jobs:
  deploy:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Build bundle
        run: |
          cd transformations
          zip -r ../bundle.zip schemas/ transformations/

      - name: Validate (dry run)
        run: |
          STATUS=$(curl -s -o /dev/null -w "%{http_code}" \
            -X POST \
            -H "X-Admin-Key: $UTLXE_ADMIN_KEY" \
            -F "file=@bundle.zip" \
            "$UTLXE_ADMIN_URL/admin/bundle/validate")
          if [ "$STATUS" != "200" ]; then
            echo "Validation failed with status $STATUS"
            exit 1
          fi

      - name: Deploy
        run: |
          curl -sf \
            -X POST \
            -H "X-Admin-Key: $UTLXE_ADMIN_KEY" \
            -F "file=@bundle.zip" \
            "$UTLXE_ADMIN_URL/admin/bundle"

      - name: Verify
        run: |
          for dir in test-data/*/; do
            NAME=$(basename "$dir")
            INPUT="$dir/input.json"
            EXPECTED="$dir/expected-output.xml"
            if [ -f "$INPUT" ]; then
              RESULT=$(curl -sf \
                -X POST \
                -H "X-Admin-Key: $UTLXE_ADMIN_KEY" \
                -H "Content-Type: application/json" \
                -d @"$INPUT" \
                "$UTLXE_ADMIN_URL/admin/transformations/$NAME/test")
              echo "$NAME: $RESULT"
            fi
          done
```

The workflow validates before deploying, and verifies after. If validation fails, the deployment never happens. If verification fails, the pipeline reports the error — the transformations are already deployed, but the team is notified.

9.5 Azure DevOps Pipeline

The structure is similar, using Azure CLI tasks for authentication:

```
trigger:
  branches:
    include:
      - main

pool:
  vmImage: 'ubuntu-latest'

variables:
  - group: utlxe-secrets

steps:
  - script: |
      cd transformations
      zip -r $(Build.ArtifactStagingDirectory)/bundle.zip schemas/ transformations/
    displayName: 'Build bundle'

  - script: |
      curl -sf -X POST \
        -H "X-Admin-Key: $(UTLXE_ADMIN_KEY)" \
        -F "file=@$(Build.ArtifactStagingDirectory)/bundle.zip" \
        "$(UTLXE_ADMIN_URL)/admin/bundle/validate"
    displayName: 'Validate'

  - script: |
      curl -sf -X POST \
        -H "X-Admin-Key: $(UTLXE_ADMIN_KEY)" \
        -F "file=@$(Build.ArtifactStagingDirectory)/bundle.zip" \
        "$(UTLXE_ADMIN_URL)/admin/bundle"
    displayName: 'Deploy'
```

Store `UTLXE_ADMIN_KEY` and `UTLXE_ADMIN_URL` in an Azure DevOps variable group marked as secret.

9.6 Rollback

Rollback is straightforward: re-deploy a previous bundle. Keep previous bundles as pipeline artifacts or in a git tag.

```
# Download previous bundle from artifact storage
# Re-deploy it
curl -X POST -H "X-Admin-Key: $KEY" \
  -F "file=@bundle-v1.2.3.zip" \
  http://<admin>:8081/admin/bundle
```

The bundle upload is atomic — the old transformations are replaced in one operation. There is no partial state.

Alternatively, use the export endpoint to create a backup before deploying:

```
# Backup current state
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/bundle -o backup-before-deploy.zip
```

```
# Deploy new version
curl -X POST -H "X-Admin-Key: $KEY" \
  -F "file=@new-bundle.zip" \
  http://<admin>:8081/admin/bundle

# If something goes wrong, rollback
curl -X POST -H "X-Admin-Key: $KEY" \
  -F "file=@backup-before-deploy.zip" \
  http://<admin>:8081/admin/bundle
```

9.7 Multi-Environment Promotion

Use the same bundle across environments with environment-specific configuration:

Environment	Validation policy	How
Development	`off`	Runtime override via Admin API
Staging	`warn`	`transform.yaml` config
Production	`strict`	`transform.yaml` config

The `.utlx` source and schemas are identical across environments. Only the `transform.yaml` varies (or the runtime override via the Admin API).

9.8 The Full Cycle

A commit to `main` triggers the pipeline. Within two minutes:

1. The bundle is built from the repository.
2. It is validated against the running UTLXe (dry run — does it compile?).
3. It is deployed (atomic replacement of all transformations).
4. Each transformation is tested with sample input.
5. The pipeline reports success or failure.

From commit to verified production deployment in under two minutes, with no custom Docker images, no container restarts, and no downtime.

10 Troubleshooting

This chapter lists the most common problems, their symptoms, and how to fix them.

10.1 Container Starts But `ready=false`

Symptom: Health endpoint returns `{"status":"UP", "transformations":0, "ready":false}`. No traffic is routed.

Cause: No transformations are loaded. The container started empty.

Fix:

- If using persistent storage: check that the Azure File Share is mounted correctly. Run `az containerapp show` and verify the volume mount configuration.
- If using CI/CD re-deploy: check that the pipeline ran and uploaded the bundle.
- Manual fix: upload a transformation via the Admin API.

10.2 Transformation Upload Fails (400)

Symptom: `POST /admin/transformations/{name}` returns 400 with an error message.

Cause: Syntax error in the `.utlx` source. The compilation failed.

Fix: Read the error response — it includes the line number and a description of the problem.

```
{
  "status": "rejected",
  "errors": [
    {"transformation": "invoice", "line": 14, "message": "Unknown function: concatX"}
  ]
}
```

Test locally before uploading: `echo '{}' | utlx -e 'your expression'` catches syntax errors immediately.

10.3 Messages Failing (500 on Data Plane)

Symptom: The data plane returns 500 for some or all messages. Error count increases in Prometheus.

Cause: Runtime error in the transformation — null reference, type mismatch, missing field.

Diagnose:

```
curl -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/transformations/invoice-to-ubl/errors
```

This shows the last 100 errors with the input that caused each failure.

Fix: Upload a corrected transformation. Test with the failing input before resuming:

```
curl -X POST -H "X-Admin-Key: $KEY" \
  -d '{"the":"failing input"}' \
  http://<admin>:8081/admin/transformations/invoice-to-ubl/test
```

10.4 High Latency

Symptom: p99 latency exceeds 100ms or response times are inconsistent.

Cause 1 — Large messages: Parsing and serialization dominate for messages above 50 KB.

Check average message size and consider whether the transformation can be simplified, or upgrade to the Professional plan for more memory.

Cause 2 — GC pressure: The heap is too small for the workload.

Check `utlxe_heap_used_bytes` in Prometheus. If it stays above 80% of the max, upgrade the plan.

Cause 3 — Swap: The container has swap enabled.

Fix: set memory request equal to memory limit in the Container App configuration. See Chapter 7 for details.

10.5 Out of Memory (OOM Kill)

Symptom: Container restarts unexpectedly. Container logs show `Killed` or the exit code is 137.

Cause: A message (or burst of messages) consumed more heap than available. The JVM exceeded the container memory limit, and Kubernetes killed the process.

Fix:
- Set `maxInputSize` to reject oversized messages before parsing.
- Upgrade to a larger plan.
- Check for transformations that create large intermediate objects (deeply nested `map` of `map` operations).

10.6 Admin API Returns 403

Symptom: All admin endpoints return 403 Forbidden.

Cause 1: `UTLXE_ADMIN_KEY` is not set. When no key is configured, all admin endpoints are locked.

Cause 2: The `X-Admin-Key` header value does not match the environment variable.

Fix: Verify the key:

```
# Check if the env var is set
az containerapp show -n utlxe -g myResourceGroup \
  --query "properties.template.containers[0].env"
```

10.7 Transformations Lost After Restart

Symptom: After a container restart, health shows `transformations: 0`.

Cause: No persistent storage is configured. The container filesystem is ephemeral.

Fix:
- Enable persistent storage (Azure Files volume mount) by redeploying with the "Enable persistent transformation storage" option.
- Or set up a CI/CD pipeline to re-deploy the bundle after each container start.

10.8 Dapr Not Delivering Messages

Symptom: Messages are in the Service Bus queue but UTLXe is not processing them.

Cause 1 — Not ready: UTLXe health shows `ready: false`. Dapr waits for the app to be ready before delivering messages.

Fix: Upload transformations. Dapr begins delivering once `ready: true`.

Cause 2 — Binding name mismatch: The Dapr component name does not match a transformation name, and no `X-UTLXe-Transform` header is configured.

Fix: Verify that the Dapr component `metadata.name` matches the transformation name, or configure the transform header override.

Cause 3 — Transformation is paused: UTLXe returns 503, Dapr does not acknowledge.

Fix: Resume the transformation:

```
curl -X POST -H "X-Admin-Key: $KEY" \
  http://<admin>:8081/admin/transformations/{name}/resume
```

10.9 Schema Validation Failing Unexpectedly

Symptom: Messages that used to work are now rejected by validation.

Cause: The schema was updated but the incoming messages still use the old format.

Quick fix: Disable validation temporarily:

```
curl -X POST -H "X-Admin-Key: $KEY" \
  -d '{"policy":"off"}' \
  http://<admin>:8081/admin/transformations/{name}/validation
```

Proper fix: Update the transformation to handle both the old and new message formats, or coordinate the schema change with the message producer.

10.10 Diagnostic Commands Reference

```
# Health and readiness
curl http://<internal>:8081/health

# Engine info (version, uptime, config)
curl -H "X-Admin-Key: $KEY" http://<internal>:8081/admin/info

# List transformations with status
curl -H "X-Admin-Key: $KEY" http://<internal>:8081/admin/transformations

# Recent errors for a transformation
curl -H "X-Admin-Key: $KEY" \
  http://<internal>:8081/admin/transformations/{name}/errors

# Test a transformation
curl -X POST -H "X-Admin-Key: $KEY" \
  -d '{"test":"data"}' \
  http://<internal>:8081/admin/transformations/{name}/test

# Effective validation state
curl -H "X-Admin-Key: $KEY" \
  http://<internal>:8081/admin/transformations/{name}/validation

# Container logs
az containerapp logs show -n utlxe -g myResourceGroup --follow
```

Appendices

Reference Material

11 Appendix A: API Reference

This appendix lists every endpoint on both ports with request format, response format, and status codes.

11.1 Authentication

All `/admin/*` endpoints require the header `X-Admin-Key: {value}`. The value must match the `UTLXE_ADMIN_KEY` environment variable. If the variable is not set, all admin endpoints return 403.

The `/health`, `/metrics`, and `/api/*` endpoints do not require authentication.

11.2 Admin Port (8081)

11.2.1 Bundle Endpoints

Method	Path	Description
`POST`	`/admin/bundle`	Upload a `.zip` bundle. Replaces all transformations and schemas atomically.
`GET`	`/admin/bundle`	Export the current state as a downloadable `.zip`.
`DELETE`	`/admin/bundle`	Remove all transformations and schemas. Returns to empty state.
`POST`	`/admin/bundle/validate`	Upload and validate without deploying (dry run).

11.2.2 Transformation Endpoints

Method	Path	Description
`GET`	`/admin/transformations`	List all deployed transformations with status and metrics.
`GET`	`/admin/transformations/{name}`	Get details: config, compile status, metrics, source.
`POST`	`/admin/transformations/{name}`	Deploy or update. Accepts `multipart/form-data` with `source` (required) and `config` (optional).
`POST`	`/admin/transformations/{name}/config`	Update config only. No recompile.
`DELETE`	`/admin/transformations/{name}`	Remove a single transformation.

11.2.3 Testing Endpoint

Method	Path	Description
`POST`	`/admin/transformations/{name}/test`	Execute with sample input. Returns output or error. Not counted in metrics.

11.2.4 Operational Endpoints

Method	Path	Description
`GET`	`/admin/info`	Engine version, uptime, config, persistence mode, transformation count.
`POST`	`/admin/transformations/{name}/pause`	Stop processing. Data plane returns 503. Transformation stays compiled.
`POST`	`/admin/transformations/{name}/resume`	Resume processing.
`GET`	`/admin/transformations/{name}/errors`	Recent errors (ring buffer, last 100). Accepts `?limit=N`.
`GET`	`/admin/config`	View current engine configuration.

`POST`	`/admin/config`	Update runtime-safe configuration fields.

11.2.5 Validation Override Endpoints

Method	Path	Description
`GET`	`/admin/transformations/{name}/validation`	Get effective validation state (policy, source, config default).
`POST`	`/admin/transformations/{name}/validation`	Set a runtime override. Body: `{"policy":"off"}`. Ephemeral — not persisted.
`DELETE`	`/admin/transformations/{name}/validation`	Remove override. Revert to config or header default.

11.2.6 Schema Endpoints

Method	Path	Description
`GET`	`/admin/schemas`	List all uploaded schemas with filename and size.
`GET`	`/admin/schemas/{filename}`	Download a schema file.
`POST`	`/admin/schemas/{filename}`	Upload or replace a schema. Accepts `multipart/form-data` with `file`.
`DELETE`	`/admin/schemas/{filename}`	Remove a schema.

11.2.7 Health Endpoints (no authentication)

Method	Path	Description
`GET`	`/health`	Returns status, transformation count, and ready flag.
`GET`	`/metrics`	Prometheus metrics in text exposition format.

11.3 Data Plane Port (8085)

Method	Path	Description
`GET`	`/api/transformations`	List available transformations (discovery). No authentication.
`POST`	`/api/transform/{name}`	Execute a transformation. Body is the input message. Content-Type header determines format.

11.4 Response Status Codes

Code	Meaning
200	Success.
400	Bad request — compilation error, invalid input, malformed ZIP.
403	Forbidden — missing or wrong `X-Admin-Key`, or key not configured.
404	Transformation not found.
413	Message too large — exceeds `maxInputSize`.
500	Transformation runtime error (null reference, type mismatch, etc.).
503	Transformation is paused.

12 Appendix B: Configuration Reference

12.1 Environment Variables

Variable	Default	Description
`UTLXE_ADMIN_KEY`	*(none)*	Admin API authentication key. Required — if not set, all admin endpoints return 403.
`UTLXE_HEAP_SIZE`	`1536m`	JVM heap size. Starter: `1536m`. Professional: `3072m`.
`JAVA_OPTS`	*(see below)*	Additional JVM options. Default includes G1GC, container support, AlwaysPreTouch.

Default `JAVA_OPTS`:

```
-XX:+UseG1GC -XX:MaxGCPauseMillis=200 -XX:+UseContainerSupport -XX:+AlwaysPreTouch
```

12.2 Engine Configuration (engine.yaml)

The engine configuration file is optional. If present in the bundle, it overrides the defaults.

Field	Default	Description
`maxInputSize`	`5MB`	Maximum message size before rejection. Messages larger than this are rejected with 413.
`workers`	CPU cores	Worker thread pool size.
`healthPort`	`8081`	Health + admin API port.
`dataPort`	`8085`	Data plane port.

12.3 Transformation Configuration (transform.yaml)

Per-transformation configuration. Optional — if absent, defaults apply.

Field	Default	Description
`strategy`	`COMPILED`	Execution strategy: `TEMPLATE`, `COPY`, `COMPILED`, `AUTO`.
`validationPolicy`	`strict`	Input validation: `strict` (reject), `warn` (log), `off` (skip).
`maxConcurrent`	*(unlimited)*	Maximum concurrent executions for this transformation.
`outputBinding`	*(none)*	Dapr output binding name for sending transformed messages.

Inputs and schemas:

```yaml
strategy: COMPILED
validationPolicy: strict
maxConcurrent: 4
outputBinding: orders-out
inputs:
  - name: input
    schema: order.xsd
output:
  schema: invoice.xsd
```

12.4 JVM Tuning Reference

Flag	Purpose

`-XX:+UseG1GC`	G1 garbage collector. Good for large heaps with pause-time targets.
`-XX:MaxGCPauseMillis=200`	Target maximum GC pause. G1 adjusts its behavior to meet this target.
`-XX:+UseContainerSupport`	Respect cgroup memory limits instead of reading host memory.
`-XX:+AlwaysPreTouch`	Allocate the entire heap at startup. Fail fast if not enough RAM.
`-Xmx${UTLXE_HEAP_SIZE}`	Maximum heap size. Set via `UTLXE_HEAP_SIZE` environment variable.

12.5 Container Plans

Plan	Container RAM	Heap	Max message	vCPU
Starter	2 GB	1536 MB	50 KB	1–2
Professional	4 GB	3072 MB	200 KB	2–4

The heap is set to 75% of container memory. The remaining 25% covers JVM metaspace, thread stacks, native memory, and the operating system.

12.6 Bundle ZIP Format

```
bundle.zip
  schemas/                     (optional)
    order.xsd
    invoice.json
  transformations/
    {name}/
      {name}.utlx              (required)
      transform.yaml           (optional)
  engine.yaml                  (optional)
```

12.7 Data Directory Layout

The on-disk state under `/utlxe/data/`. Written by the Admin API, read at startup.

```
/utlxe/data/
  schemas/
    order.xsd
    invoice.json
  transformations/
    invoice-to-ubl/
      invoice-to-ubl.utlx
    order-enrichment/
      order-enrichment.utlx
```

If Azure Files is mounted at `/utlxe/data/`, this directory survives container restarts.

13 Appendix C: UTL-X Quick Reference

A syntax cheat sheet for the most common transformation patterns. For the complete language reference, see the companion book *UTL-X: One Language, All Formats*.

13.1 File Structure

```
%utlx 1.0
input json
output json
---
{ transformation body }
```

13.2 Format Headers

`input json`	JSON input
`input xml`	XML input
`input csv {header: true}`	CSV with header row
`input yaml`	YAML input
`output xml`	XML output
`input json {schema: "f.xsd"}`	With schema validation

13.3 Property Access

`$input.name`	Dot notation
`$input.items[0]`	Array index (0-based)
`$input.@id`	XML attribute
`$input..name`	Recursive descent

13.4 Object Construction

```
{key: value, other: value}
{name: $input.n, ...$input}     // spread operator
```

13.5 Let Bindings

```
let x = expression
```

Multiple bindings separated by newlines or commas.

13.6 Conditionals

```
if (condition) valueA else valueB

match value {
  "A" -> resultA,
  "B" -> resultB,
  _ -> defaultResult
}
```

13.7 Array Operations

`map(arr, (x) -> expr)`	Transform each element
`filter(arr, (x) -> bool)`	Keep matching elements
`reduce(arr, init, (acc, x) -> expr)`	Aggregate to single value
`find(arr, (x) -> bool)`	First matching element
`sortBy(arr, (x) -> key)`	Sort by key function
`groupBy(arr, (x) -> key)`	Group by key function
`flatMap(arr, (x) -> arr)`	Map and flatten
`size(arr)`	Array length
`first(arr)` / `last(arr)`	First or last element

13.8 Pipe Operator

```
$input.items
  | filter(_, (x) -> x.active)
  | map(_, (x) -> x.name)
  | sortBy(_, (x) -> x)
```

The underscore _ is a placeholder for the piped value.

13.9 String Functions

`concat(a, b, c)`	Concatenate
`upperCase(s)` / `lowerCase(s)`	Case conversion
`trim(s)`	Remove whitespace
`replace(s, old, new)`	Replace substring
`contains(s, sub)`	Check substring
`substring(s, start, end)`	Extract substring
`length(s)`	String length
`split(s, delimiter)`	Split to array
`startsWith(s, prefix)` / `endsWith(s, suffix)`	Check prefix or suffix

13.10 Math Functions

`round(n, decimals)`	Round to decimal places
`floor(n)` / `ceil(n)`	Floor or ceiling
`abs(n)`	Absolute value
`min(a, b)` / `max(a, b)`	Minimum or maximum

13.11 Type Functions

`toString(v)` / `toNumber(v)` / `toBoolean(v)`	Type conversion
`isNull(v)`	Null check
`typeOf(v)`	Type name as string

13.12 Date Functions

`now()`	Current timestamp
`formatDate(d, pattern)`	Format a date (e.g., `"yyyy-MM-dd"`)
`parseDate(s, pattern)`	Parse a date string

13.13 Security Functions

`sha256(s)` / `sha512(s)`	Cryptographic hash
`hmacSHA256(s, key)`	HMAC signature
`encryptAES(data, key)` / `decryptAES(data, key)`	AES encryption
`base64Encode(s)` / `base64Decode(s)`	Base64 encoding
`mask(s, start, end, char)`	Mask sensitive data
`env("VAR_NAME")`	Read environment variable (engine only)

13.14 User-Defined Functions

```
function name(param1, param2) {
  expression
}
```

Defined before the main transformation body. Can call standard library functions and other user-defined functions.